Handwriting robot

Reading handwriting and reproducing an individual writing style

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Project overview

The system reads photographed handwriting, identifies the writer and uses stored glyphs to write new text in that person's style with a two-axis pen gantry.

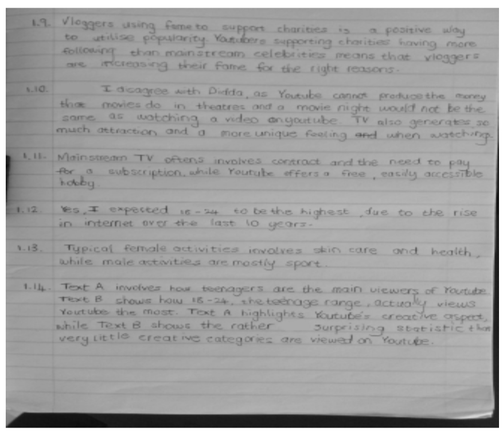
READ A PAGE: Camera image → Page processing → Text and writer

WRITE NEW TEXT: New text + writer profile → Glyph paths + G-code → Gantry and pen

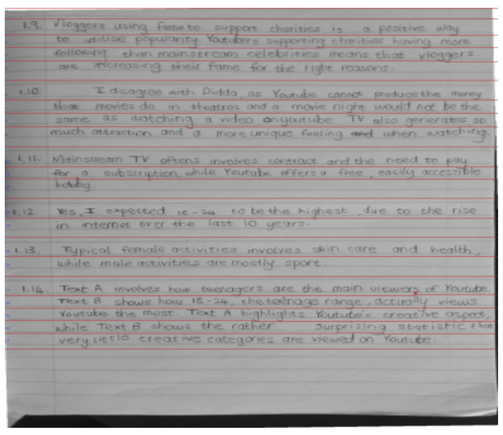
Preparation uses labelled handwriting to train the models and build the writer profiles.

01 Page processing

Photographed page



Detected line boundaries



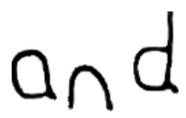
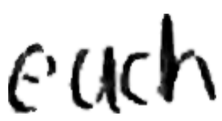
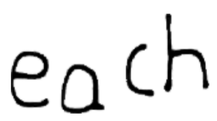
Correct uneven lighting and rotation, then separate the handwriting into ordered line images.

02 Text and writer recognition

Recognition screenshot

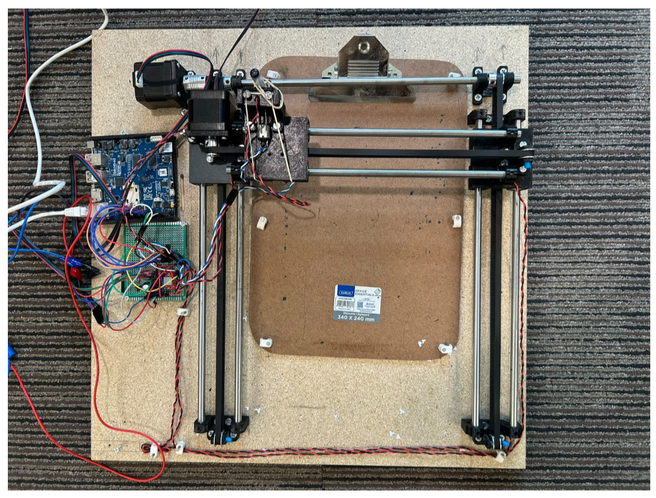
A line reader recognises the text. A separate classifier identifies the writer using normalised stroke width.

03 Handwriting reproduction

	Original handwriting	Robot handwriting
Writer 150		
Owen		
Yeukita		

Store each writer's glyph shapes, place and join them, then convert the paths into G-code for the gantry.

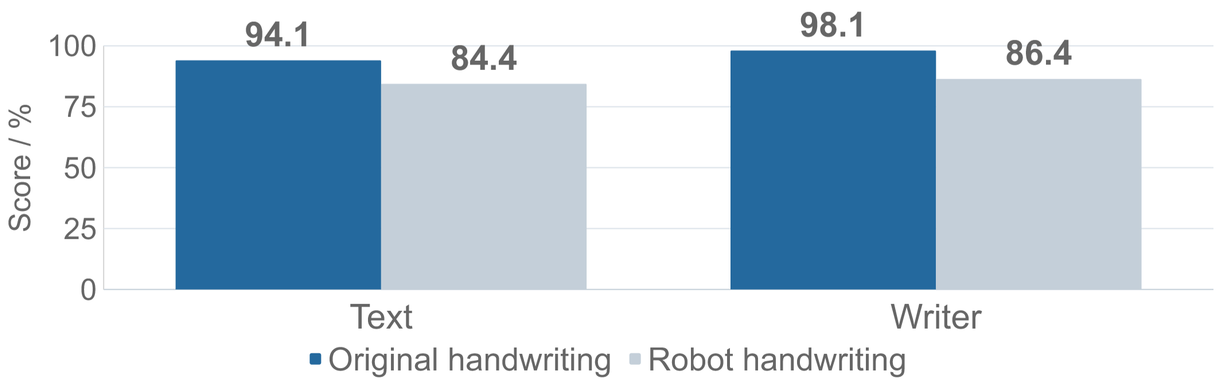
04 Gantry and pen control



Two belt-driven axes position the pen. A separate motor raises it, while switches support homing and stopping.

Results

Recognition scores



Original validation and photographed robot output are separate test sets. Physical output has 184 lines.

70 / 70
measured lengths within 0.5 mm
Largest error 0.49 mm

25 / 25
completed jobs within 4 s per character

11 writer styles
Classification time remains above target.
Connected letters need more context.

Faculty of Engineering, Built Environment and Information Technology
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Department of Electrical, Electronic and Computer Engineering

EPR400 project, 2026. Results from the project qualification records.
Length measurements assess drawn dimensions. Writing time covers completed jobs.